

25CPES102

Lecture 3

Electronic Materials and Devices

MOSFETs

Metal-oxide-semiconductor
field-effect transistors

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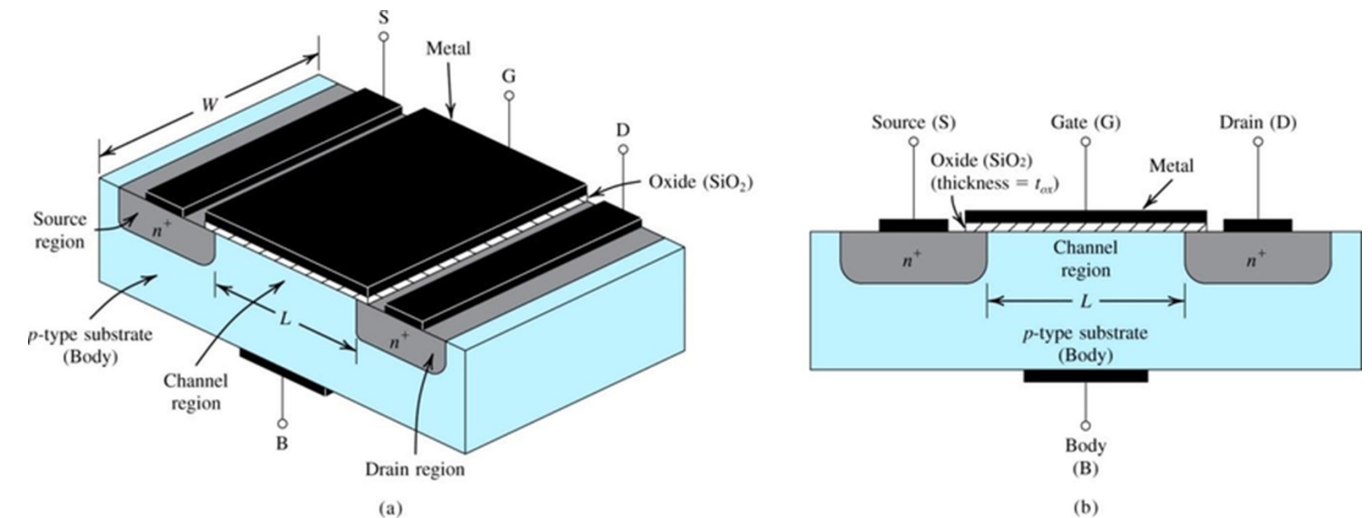
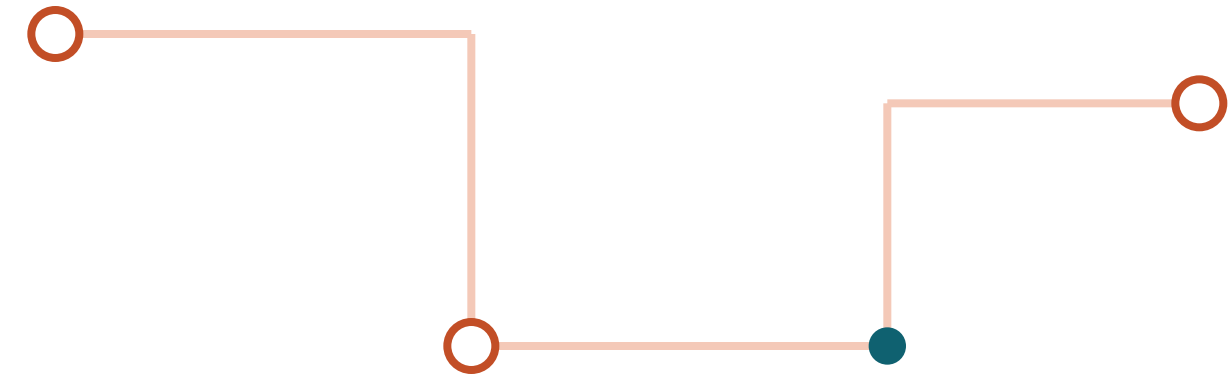


Figure: Sedra & Smith, Microelectronic Circuits

Lecture map

01

Structure and operation

how a channel forms; the threshold voltage



02

I–V characteristics

cutoff, triode and saturation



03

MOSFETs in DC circuits

finding the operating point



04

The MOSFET as an amplifier

transconductance, small-signal model



05

The MOSFET as a switch



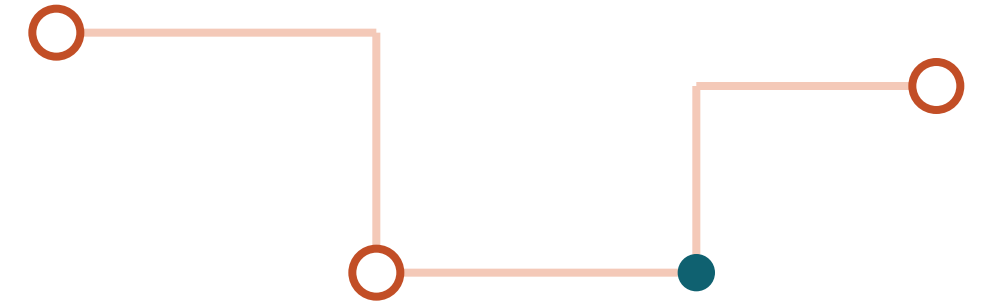
By the end you can

- Explain how v_{GS} creates a channel and how v_{DS} pinches it off
- Pick the right region and use its i_D equation
- Find the DC operating point of a MOSFET circuit
- Find g_m and the gain of a simple MOSFET amplifier

1

Structure and operation

- The device and its terminals
- No gate voltage: no current
- The threshold voltage
- Tapering and pinch-off
- CMOS; n- and p-channel symbols



1 Structure and operation

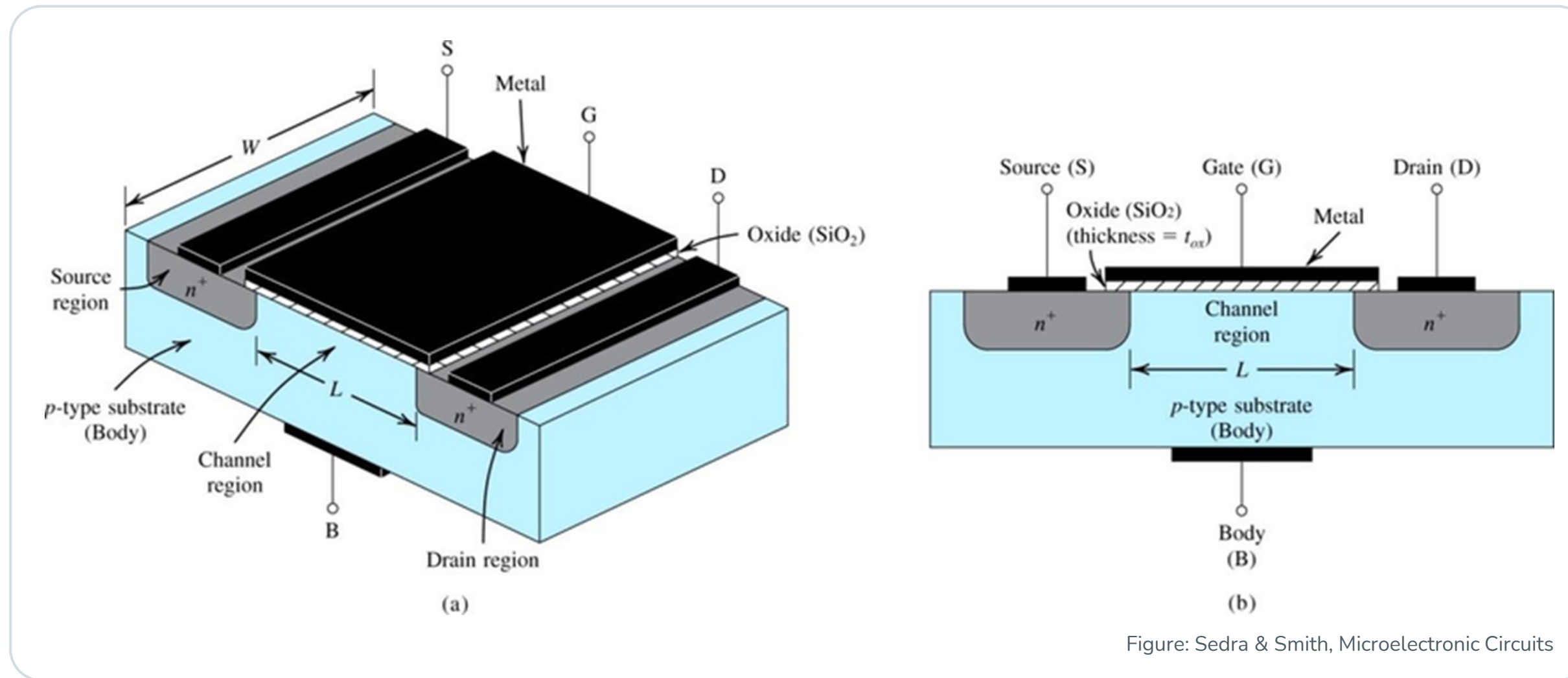
2 I-V characteristics

3 MOSFETs in DC circuits

4 The MOSFET as an amplifier

5 The MOSFET as a switch

The n-channel enhancement MOSFET



- Four terminals: **source** (S), **gate** (G), **drain** (D) and **body** (B).
- A thin oxide (SiO_2) insulates the gate, so the gate draws **no current**.
- The channel region between the n^+ source and drain has length L and width W .

With no gate voltage, no current flows

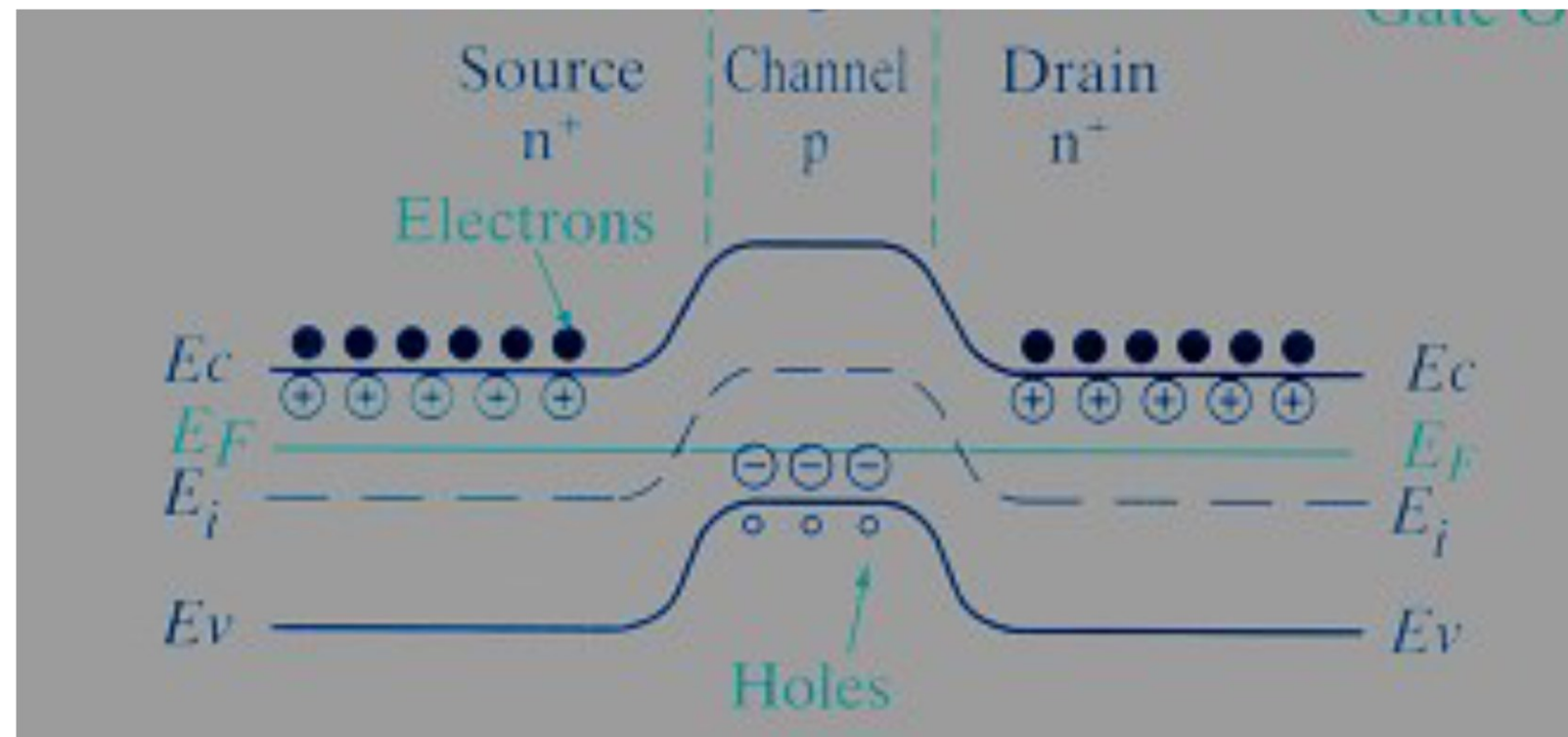


Figure: web source, via the original lecture slides

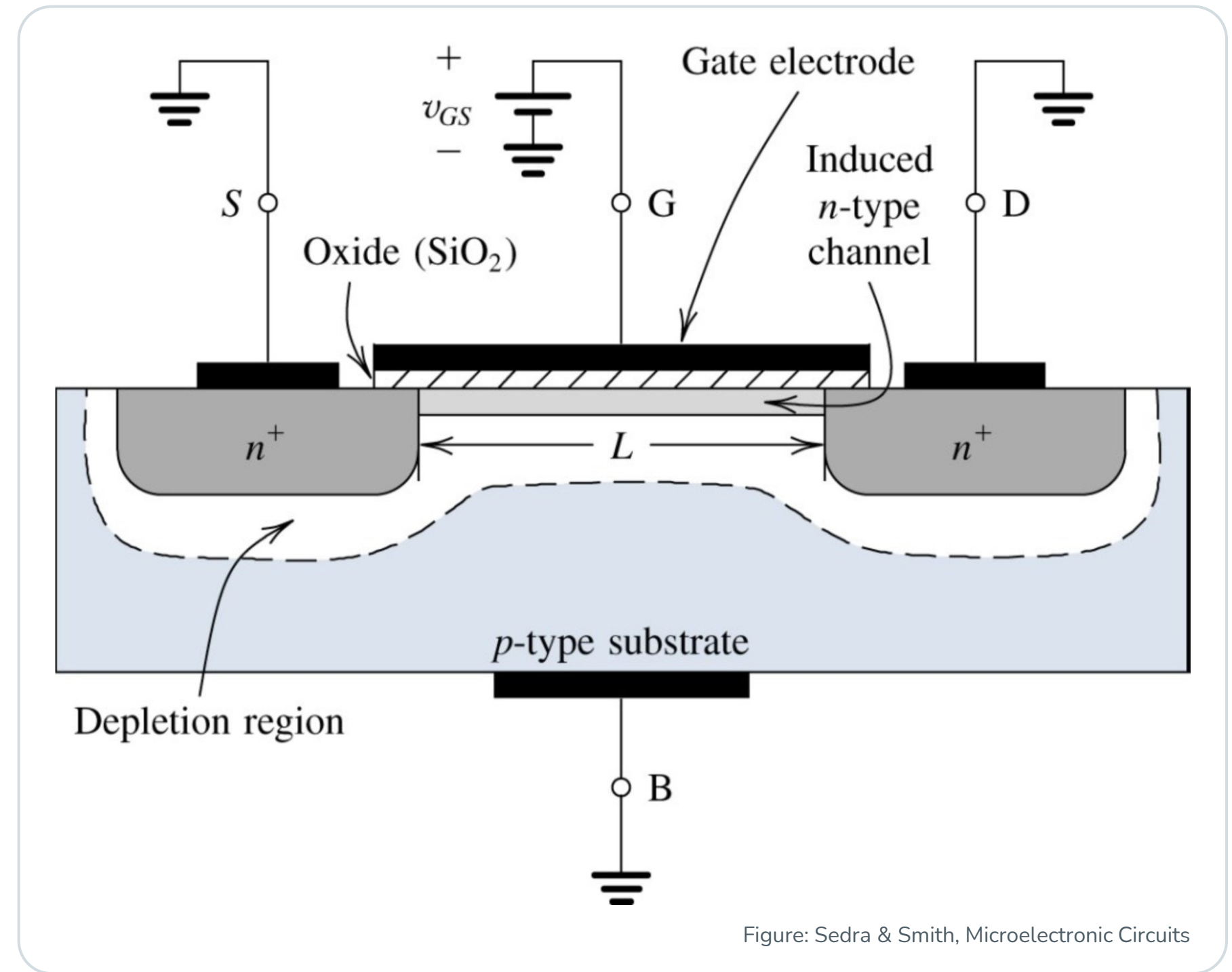
- Source and drain are n⁺, the body is p-type: two **back-to-back pn junctions** lie between source and drain.
- Whatever the polarity of v_{DS} , one junction is reverse-biased, so $i_D \approx 0$.
- The energy barrier (left) keeps the electrons of the source away from the drain.

Creating a channel: the threshold voltage

- A positive v_{GS} pushes holes away from the region under the gate and draws electrons in from the n^+ source and drain.
- Once enough electrons gather, they form an **n-type inversion layer**: the channel linking source to drain.

Definition

Threshold voltage V_t : the value of v_{GS} at which the channel just forms.



Increasing v_{DS} : the channel tapers and pinches off

- Along the channel, the voltage rises from 0 at the source to v_{DS} at the drain.
- The gate-to-channel voltage therefore falls from v_{GS} to $v_{GS} - v_{DS}$: the channel gets shallower toward the drain.

Pinch-off

When $v_{DS} = v_{GS} - V_t$, the channel depth at the drain end falls to almost zero. Increasing v_{DS} further barely changes i_D : the current **saturates**.

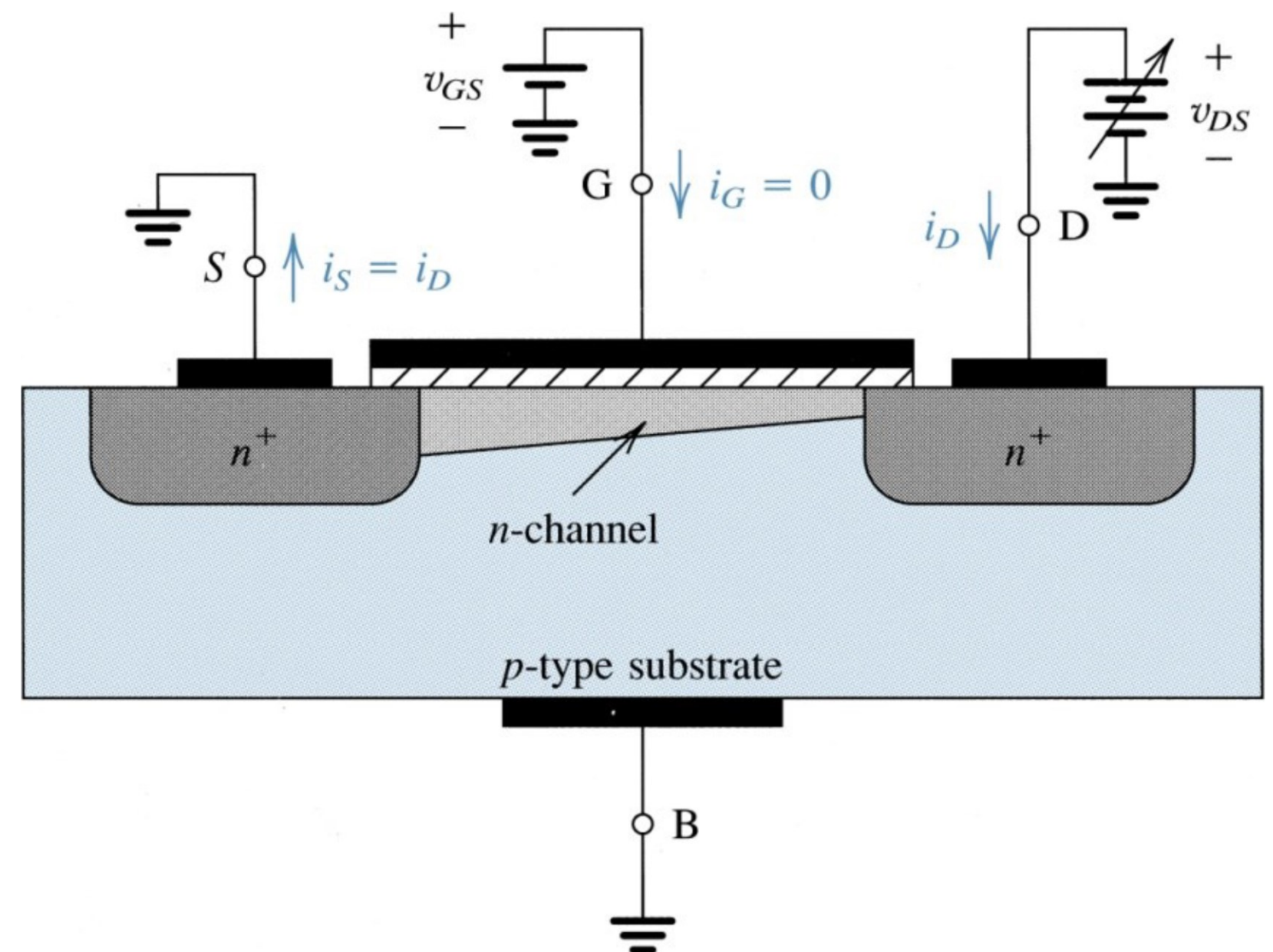
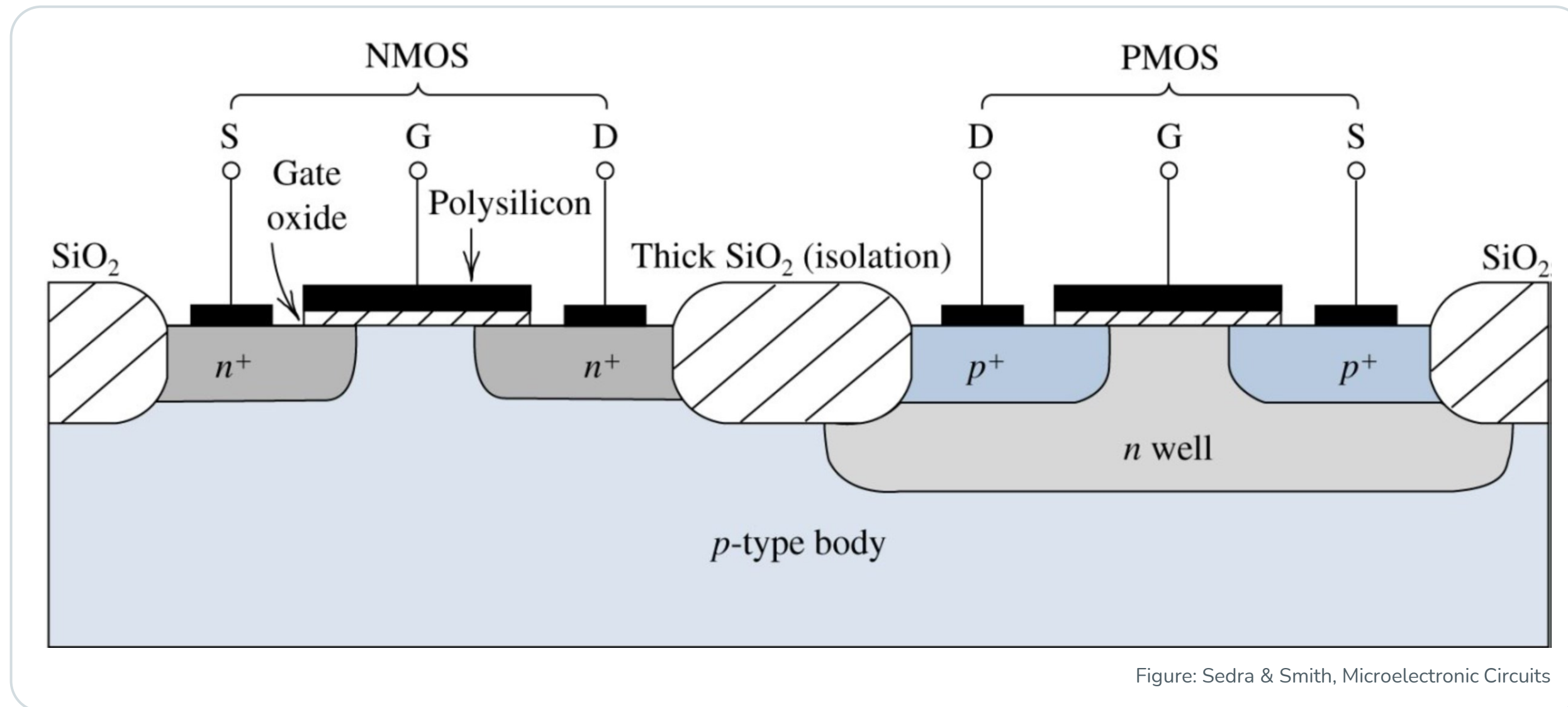


Figure: Sedra & Smith, Microelectronic Circuits

CMOS: NMOS and PMOS on one chip



- The PMOS transistor is built in an **n -well**, with p^+ source and drain.
- Thick oxide isolates neighbouring devices. CMOS circuits use both types together.

Circuit symbols: n-channel and p-channel

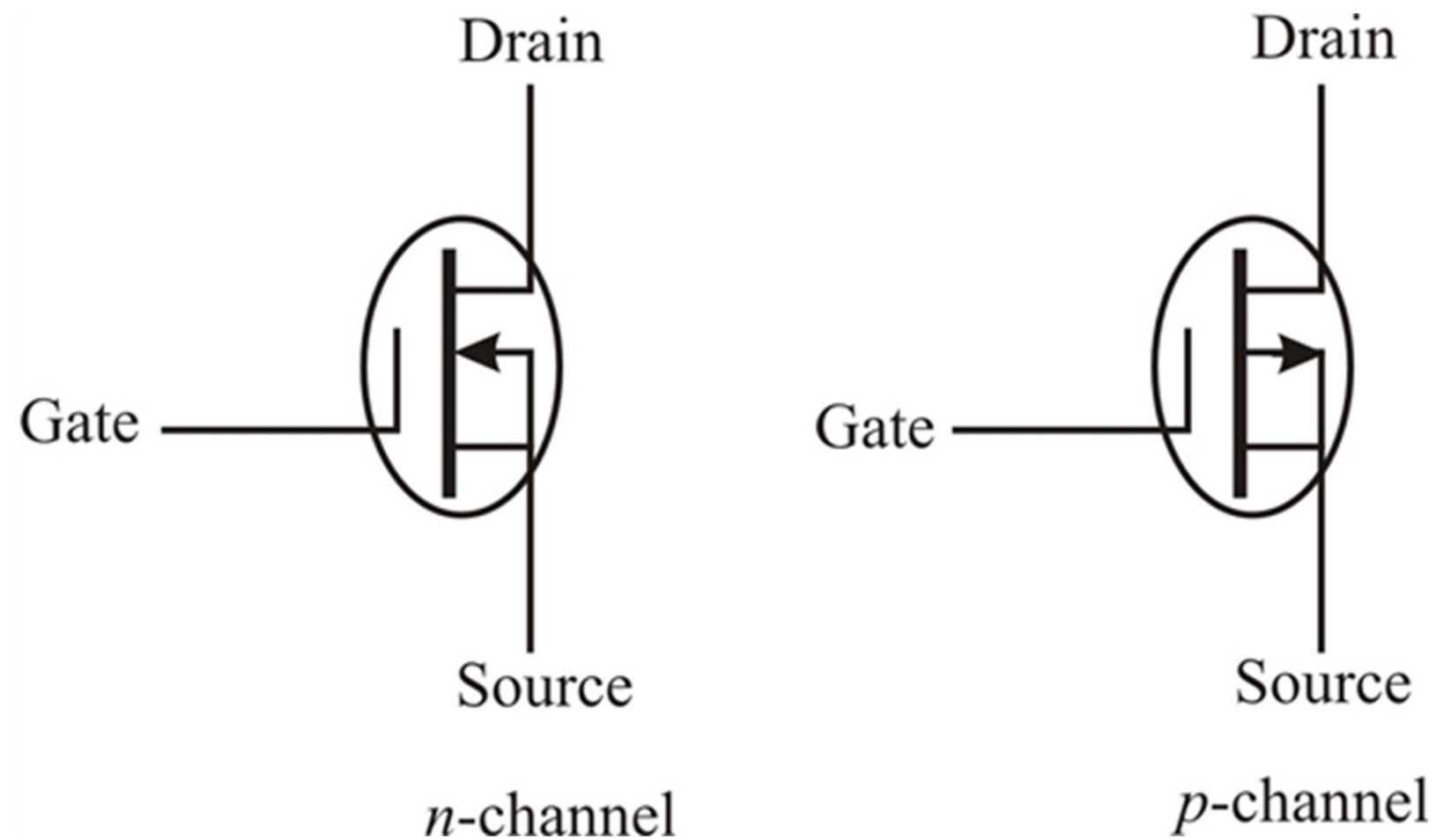


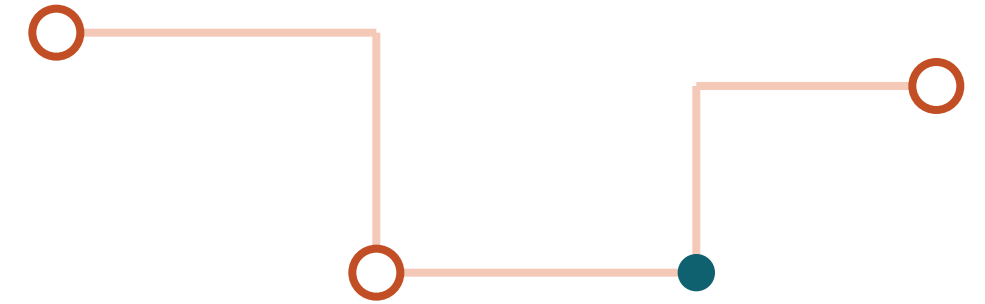
Figure: web source, via the original lecture slides

- In these symbols the arrow is on the body: it points **in** for an n-channel device and **out** for a p-channel device.
- PMOS: holes carry the current and every voltage and current is reversed. It conducts when $v_{GS} < V_t$, with V_t negative.

2

I–V characteristics

- The i_D – v_{DS} family
- Saturation: the square law
- Triode and saturation equations
- Resistor and large-signal models



1 Structure and operation

2 I–V characteristics

3 MOSFETs in DC circuits

4 The MOSFET as an amplifier

5 The MOSFET as a switch

The i_D – v_{DS} family of curves

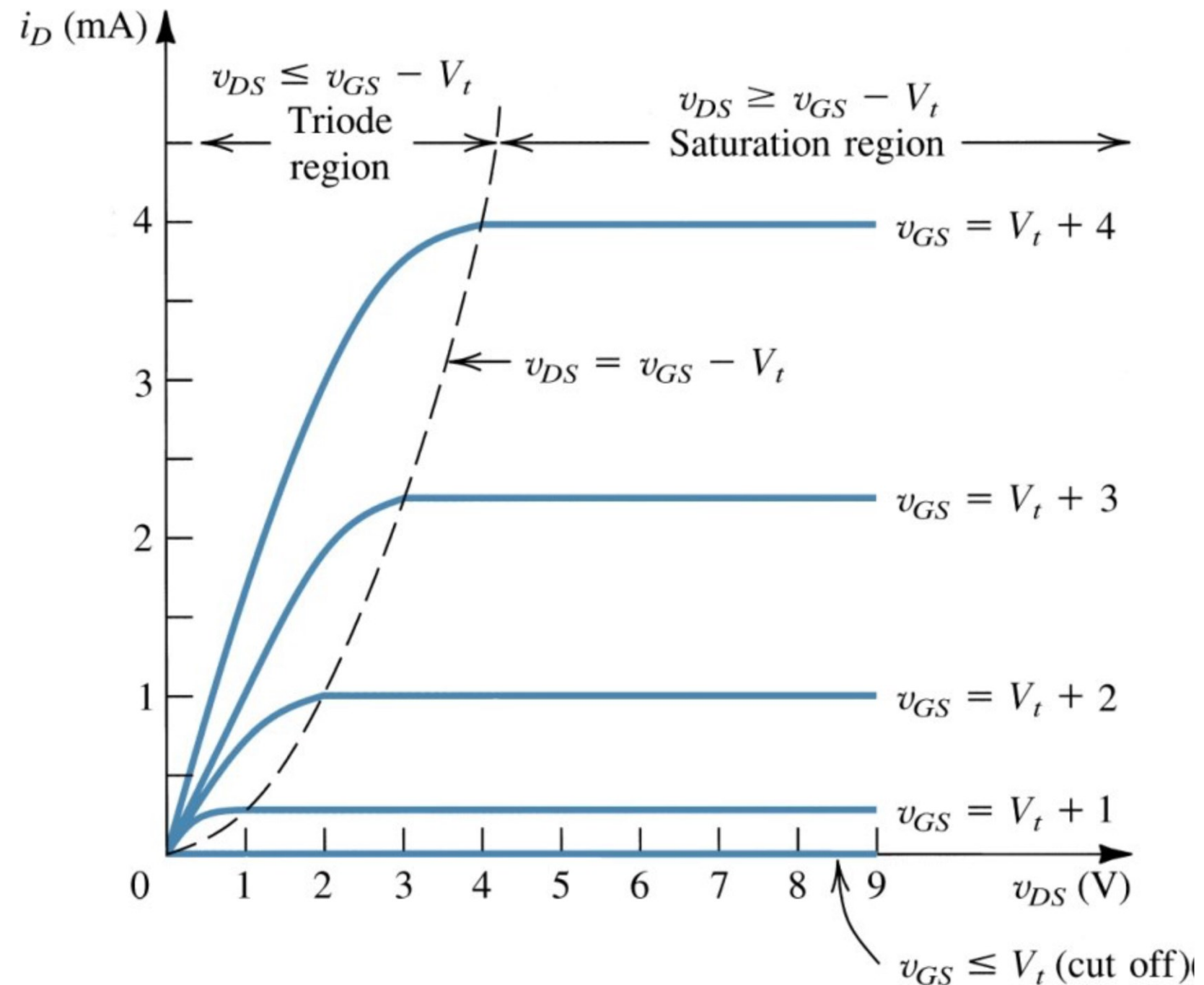


Figure: Sedra & Smith, Microelectronic Circuits

- Each curve is one value of $v_{GS} > V_t$.
- Left of the dashed line ($v_{DS} < v_{GS} - V_t$): **triode**. i_D rises with v_{DS} .
- Right of it: **saturation**. i_D is set by v_{GS} alone.
- For $v_{GS} \leq V_t$: **cutoff**, $i_D = 0$.

In saturation, i_D depends only on v_{GS}

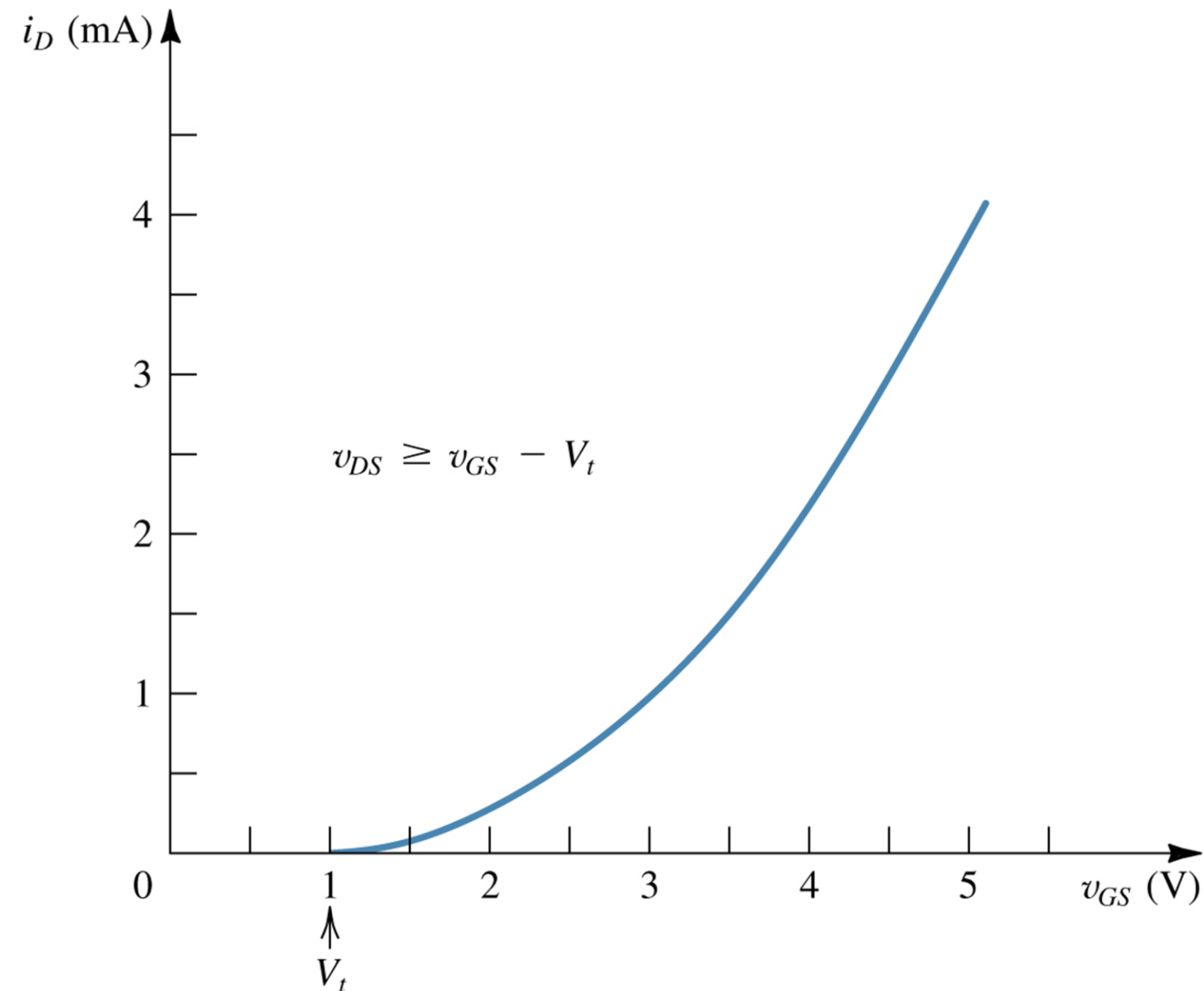


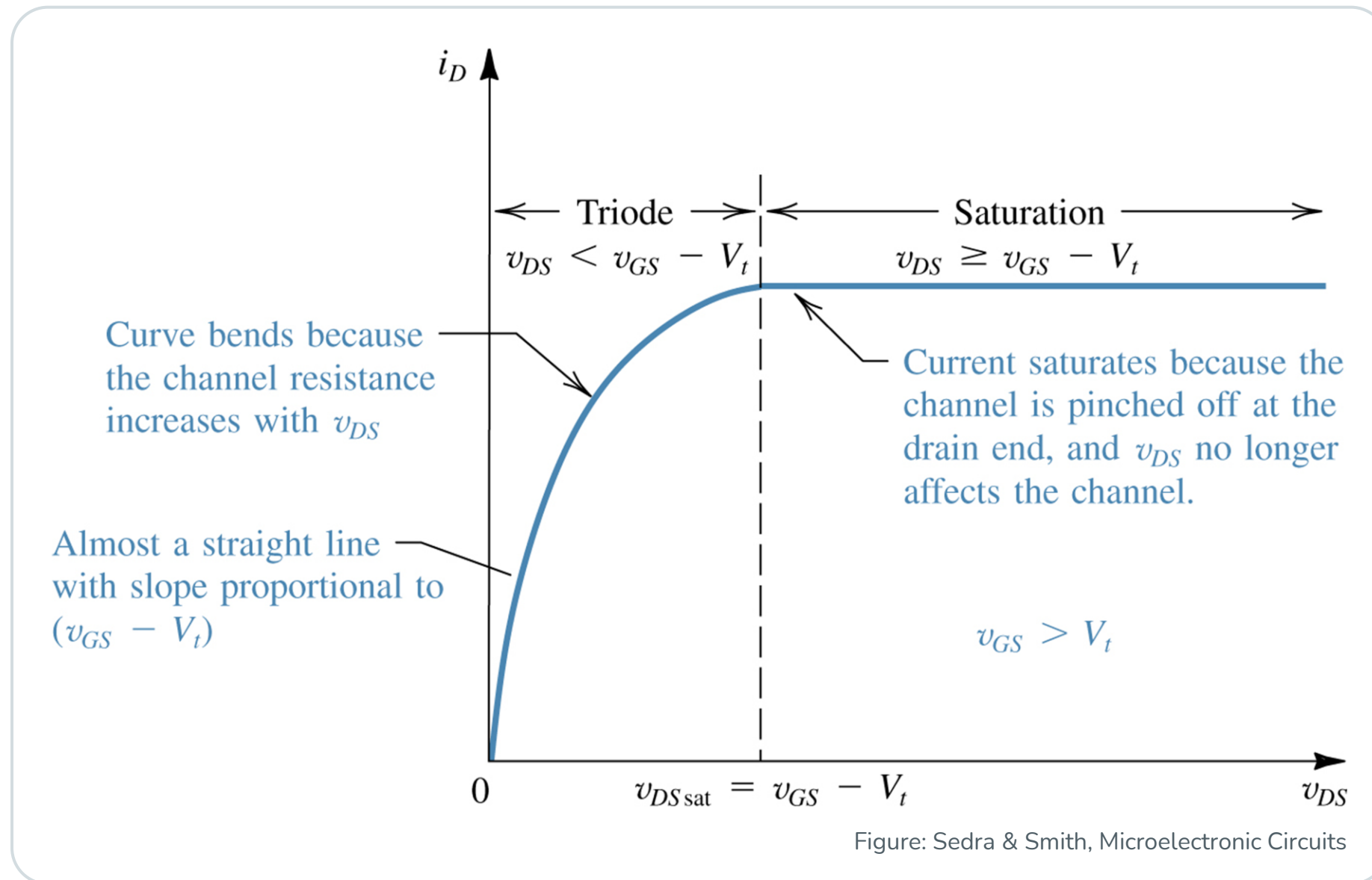
Figure: Sedra & Smith, Microelectronic Circuits

- Measured in saturation, the transfer characteristic is a parabola that starts at $v_{GS} = V_t$.

$$i_D = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2$$

- Doubling $(v_{GS} - V_t)$ quadruples i_D : the **square law**.

Triode and saturation: one curve, two equations



Triode $v_{DS} < v_{GS} - V_t$

$$i_D = k'_n \frac{W}{L} \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

Saturation $v_{DS} \geq v_{GS} - V_t$

$$i_D = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2$$

Both require $v_{GS} > V_t$: without a channel, $i_D = 0$ (cutoff).

Small v_{DS} : a voltage-controlled resistor

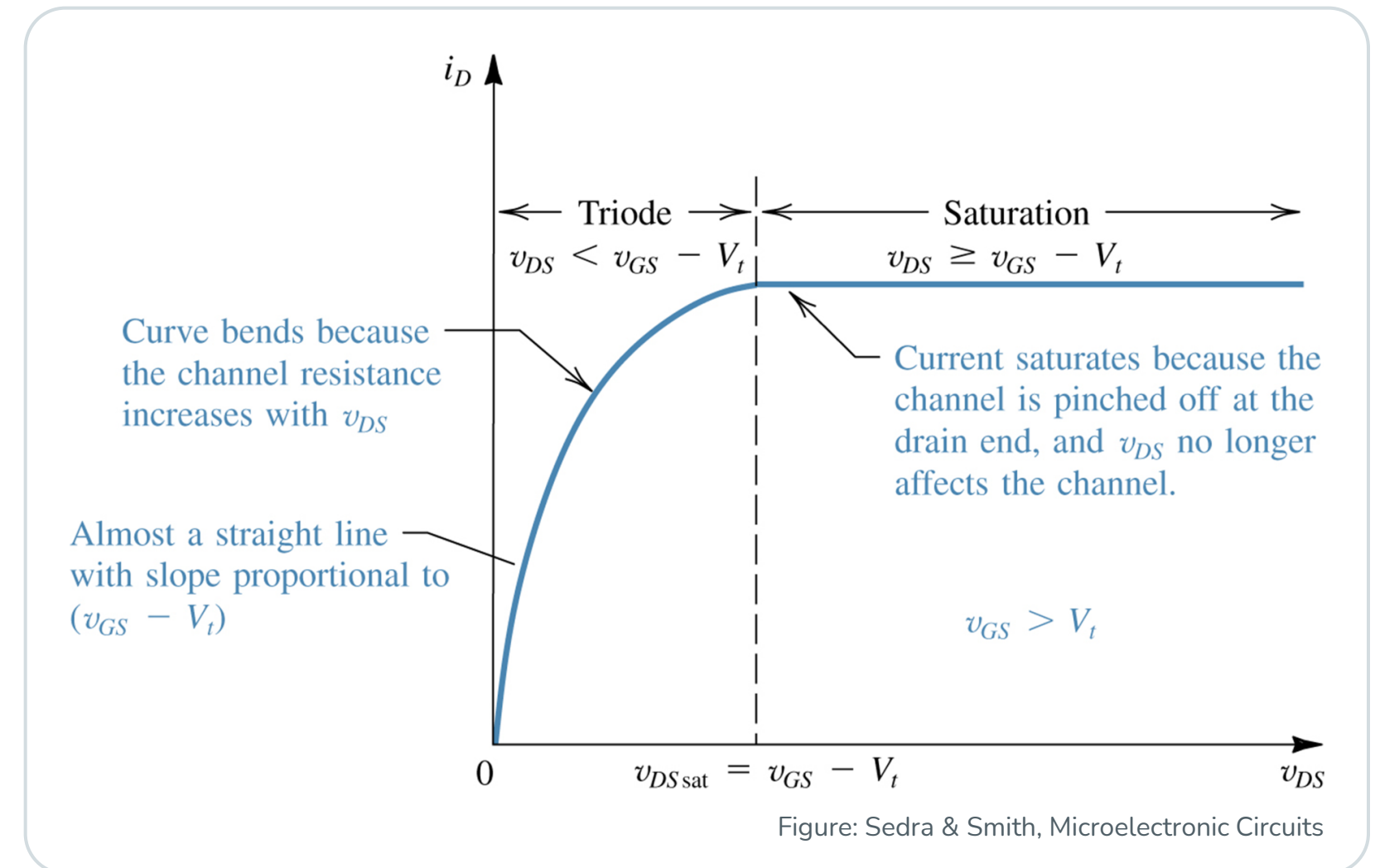
- For $v_{DS} \ll 2(v_{GS} - V_t)$ the v_{DS}^2 term is negligible:

$$i_D \approx k'_n \frac{W}{L} (v_{GS} - V_t) v_{DS}$$

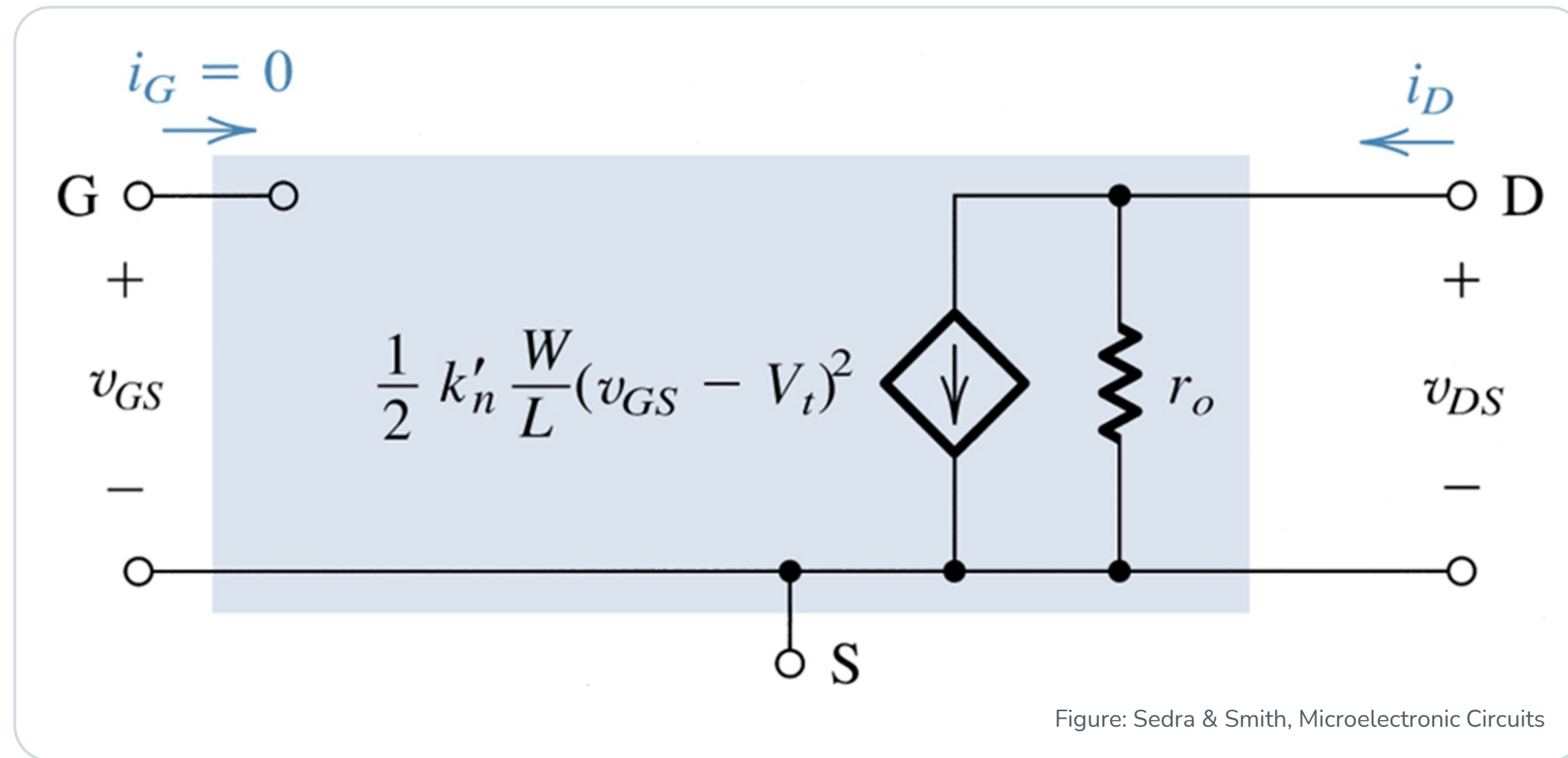
- The channel behaves as a resistor whose value v_{GS} sets:

$$r_{DS} = \left[k'_n \frac{W}{L} (v_{GS} - V_t) \right]^{-1}$$

- This is the region used when the MOSFET acts as a closed switch (Section 5).



Large-signal model in saturation



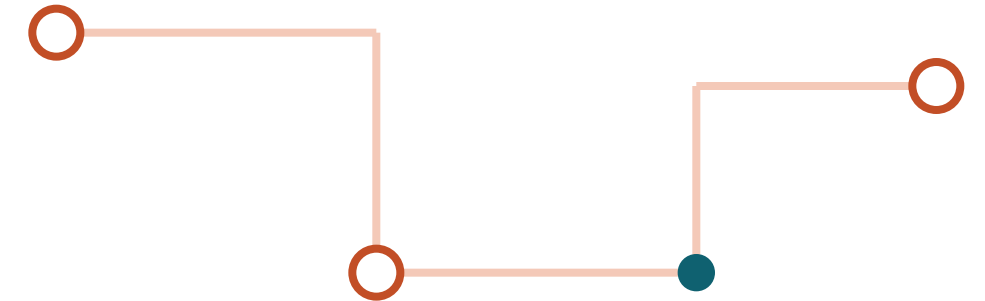
- In saturation the MOSFET acts as a **voltage-controlled current source** from drain to source.
- The gate draws no current: $i_G = 0$.

$$i_D = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2$$

3

MOSFETs in DC circuits

- Assume, solve, check
- Example: the drain voltage
- The current mirror



1 Structure and operation

2 I-V characteristics

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5 The MOSFET as a switch

DC analysis: assume, solve, check

1

Assume

a region, usually saturation (and $v_{GS} > V_t$).

2

Solve

with that region's i_D equation and KVL/KCL. Reject roots that give $v_{GS} < V_t$.

3

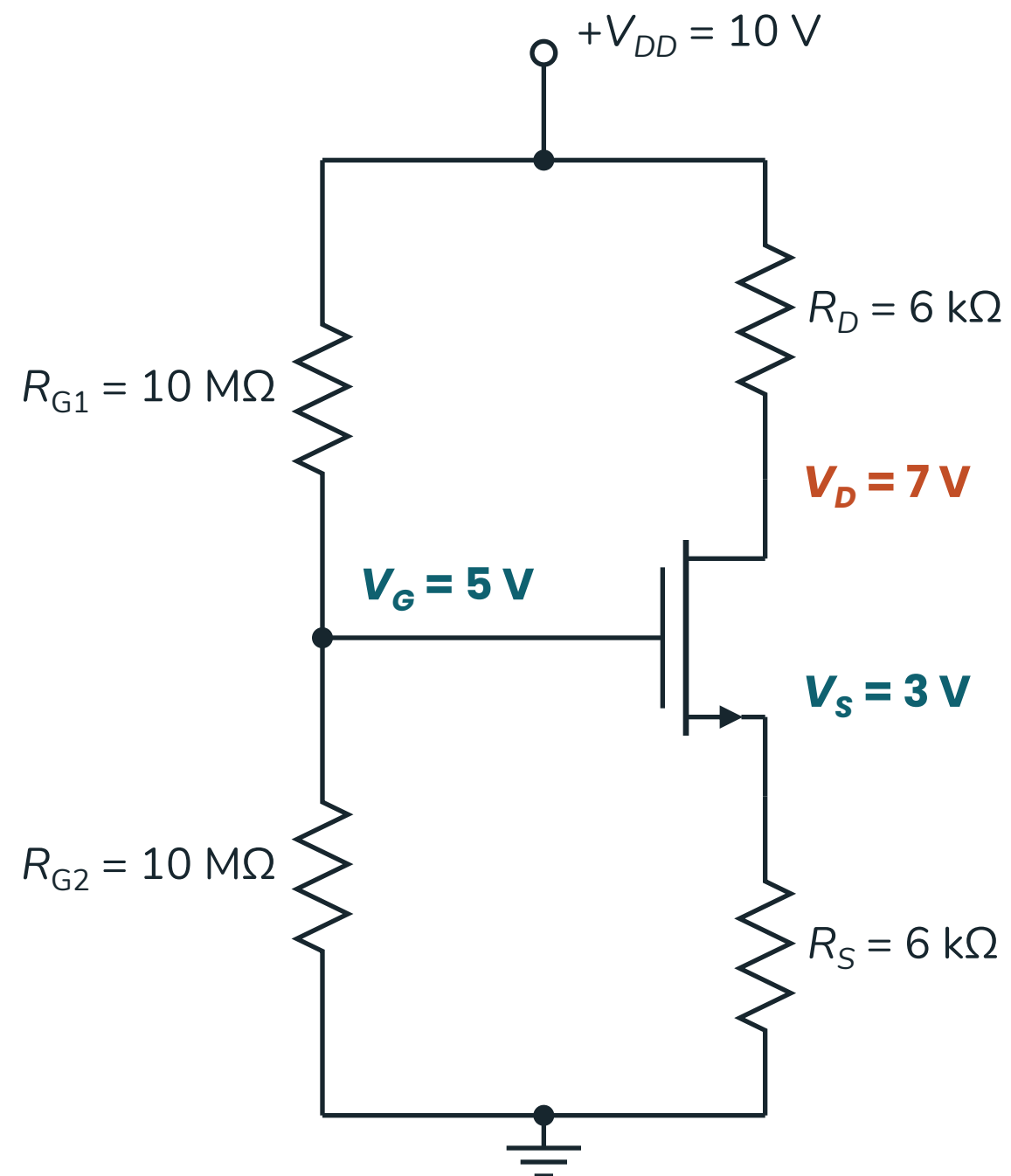
Check

the assumption: saturation needs $v_{DS} \geq v_{GS} - V_t$. If it fails, redo it in triode.

The gate draws no current, so gate-bias resistors form an unloaded divider.

Example: find the drain voltage

Given $V_t = 1 \text{ V}$ and $k'_n(W/L) = 1 \text{ mA/V}^2$. The gate draws no current.



- 1 Gate voltage: the divider is unloaded

$$V_G = V_{DD} \frac{R_{G2}}{R_{G1} + R_{G2}} = 5 \text{ V}$$
- 2 Assume saturation, with $V_{GS} = V_G - I_D R_S$ (I in mA, R in $\text{k}\Omega$)

$$I_D = \frac{1}{2} (1)(5 - 6I_D - 1)^2$$
- 3 Solve the quadratic

$$18I_D^2 - 25I_D + 8 = 0 \Rightarrow I_D = 0.89 \text{ or } 0.5 \text{ mA}$$
- 4 $I_D = 0.89 \text{ mA}$ would give $V_{GS} = -0.33 \text{ V} < V_t$ (no channel): reject.
 So $I_D = 0.5 \text{ mA}$ and $V_{GS} = 2 \text{ V}$.
- 5 Drain voltage

$$V_D = V_{DD} - I_D R_D = 10 - 0.5(6) = 7 \text{ V}$$

$V_D = 7 \text{ V}$

✓ $V_{DS} = 4 \text{ V} \geq V_{GS} - V_t = 1 \text{ V}$: saturation holds.

The current mirror

- Q_1 is diode-connected (gate tied to drain), so it is in saturation. R_{SET} sets its current I_{REF} , and with it V_{GS} .
- Q_2 has the same V_{GS} . If it is also in saturation, its current scales with W/L :

$$\frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$

- Identical transistors copy the current: $I_O = I_{REF}$.

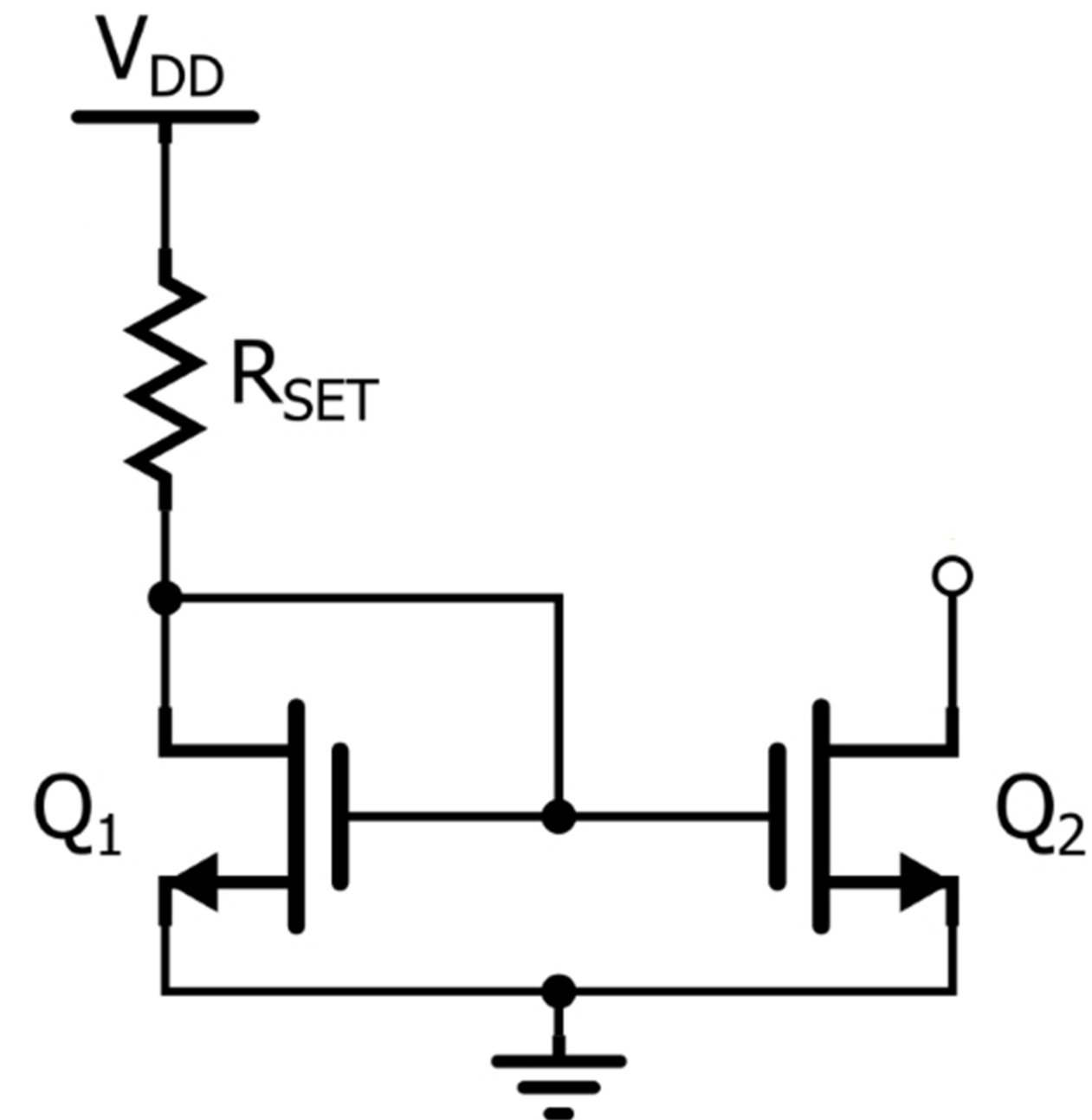
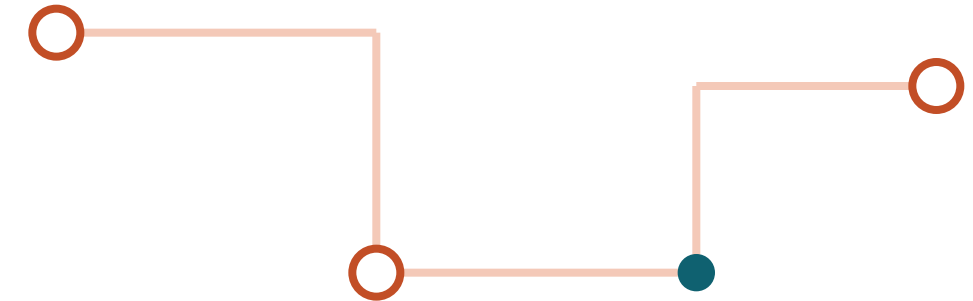


Figure: web source, via the original lecture slides

4

The MOSFET as an amplifier

- The load line
- Transconductance g_m
- Small-signal models
- Example: gain and input resistance



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5 The MOSFET as a switch

Amplifier action: the load line

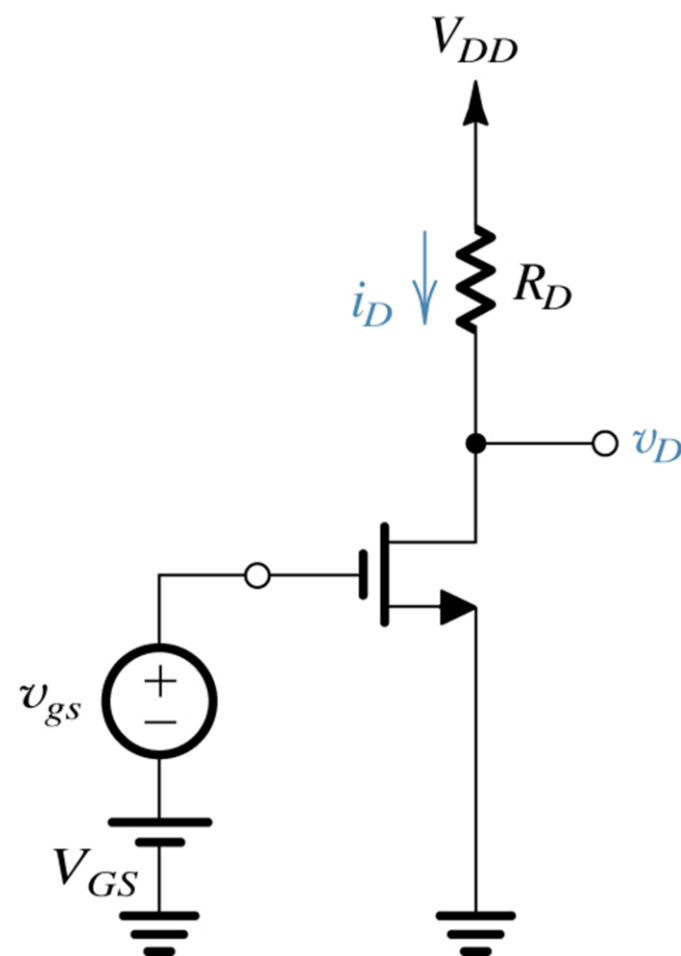


Figure: Sedra & Smith, Microelectronic Circuits

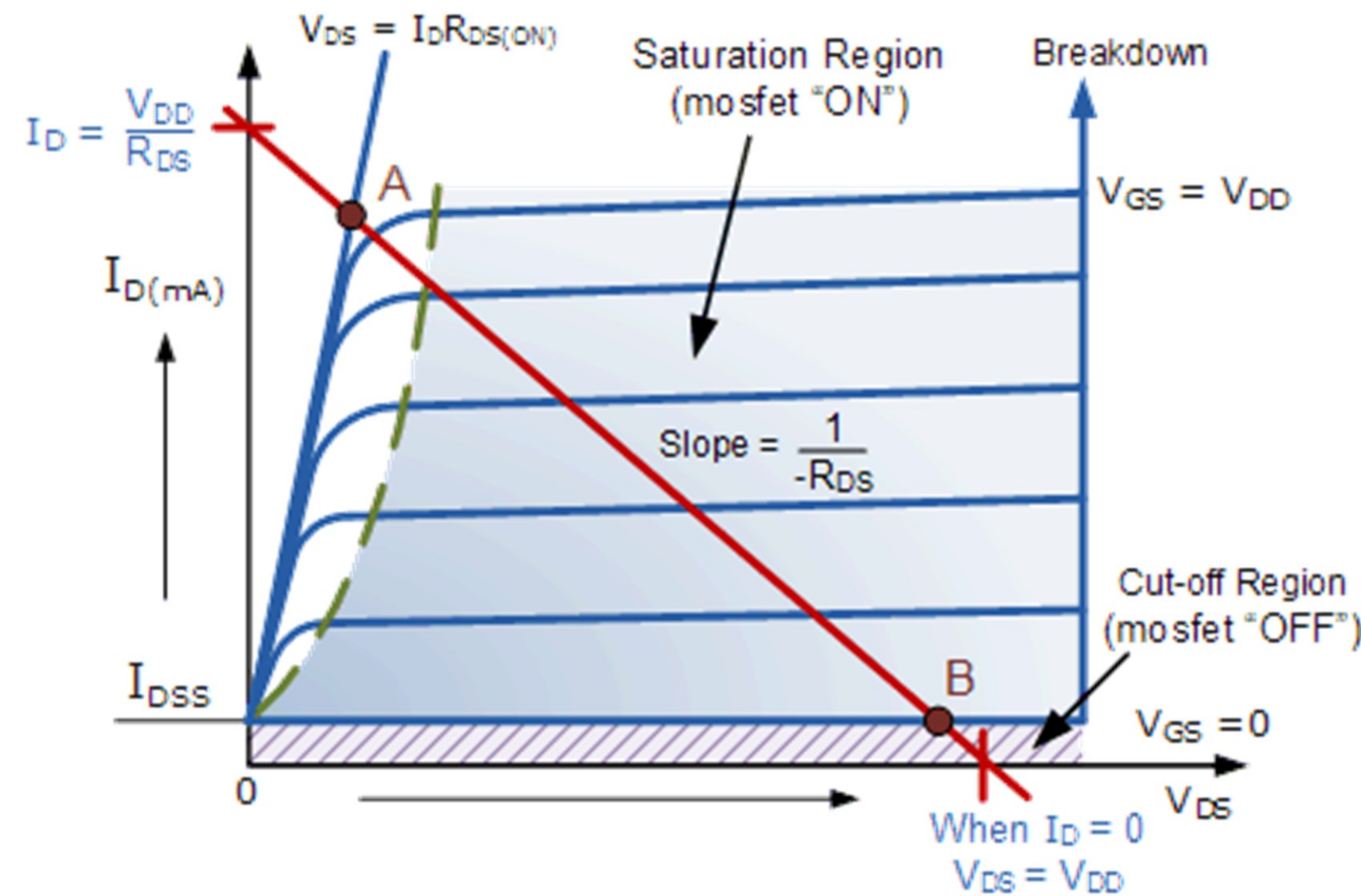


Figure: web source, via the original lecture slides

$$v_{DS} = V_{DD} - R_D i_D$$

- Bias the MOSFET in saturation, between A and B.
- A small change in v_{GS} moves the operating point along the line: a **large** change in v_{DS} .

Small-signal operation: the transconductance g_m

- 1 The gate voltage is the bias plus a small signal

$$v_{GS} = V_{GS} + v_{gs}$$

- 2 Substitute into the saturation equation

$$i_D = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_t)^2 + k'_n \frac{W}{L} (V_{GS} - V_t) v_{gs} + \frac{1}{2} k'_n \frac{W}{L} v_{gs}^2$$

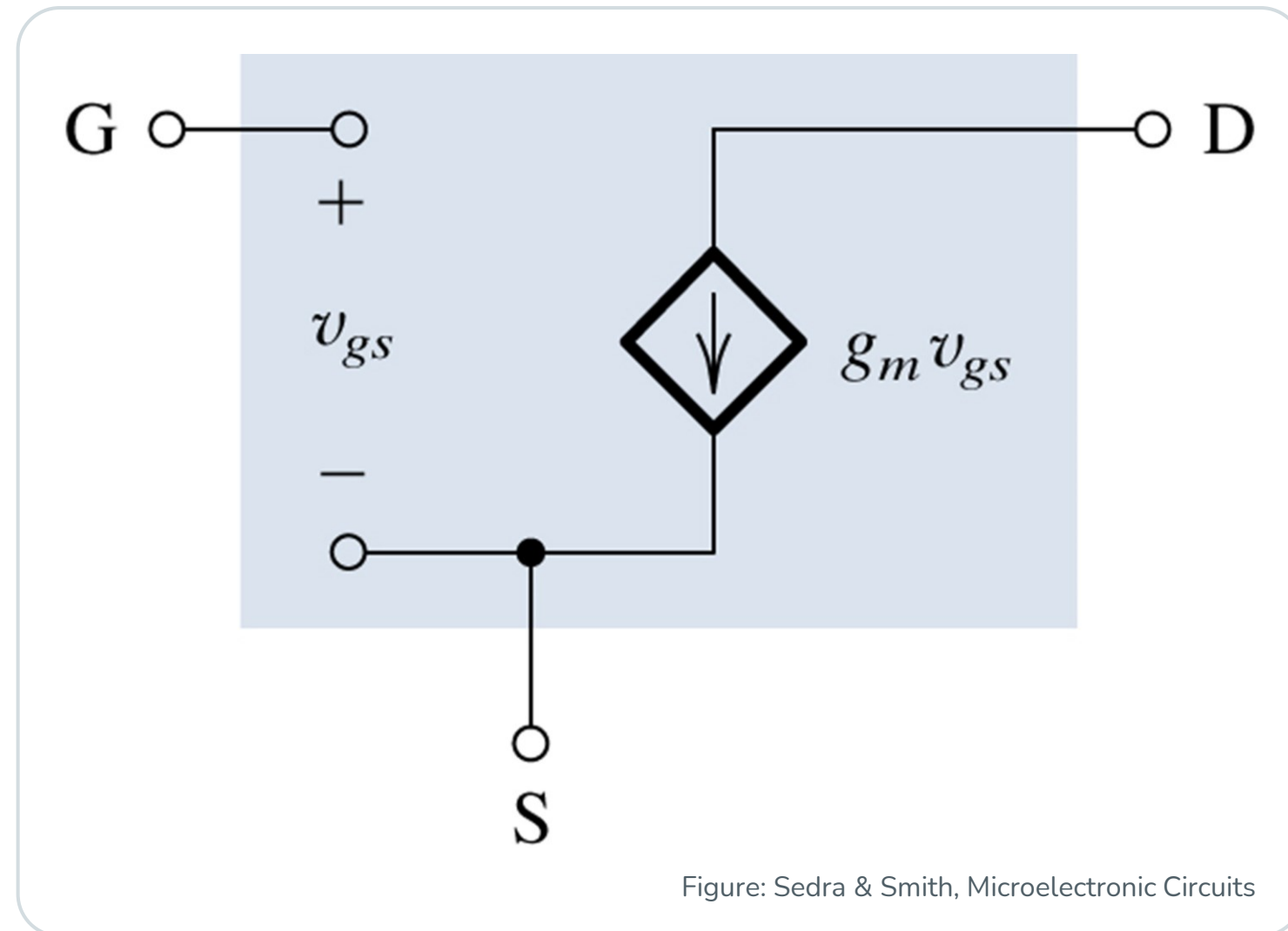
- 3 If $v_{gs} \ll 2(V_{GS} - V_t)$, the last term is negligible: the signal current is linear in v_{gs}

$$i_d = g_m v_{gs}, \quad g_m = k'_n \frac{W}{L} (V_{GS} - V_t)$$

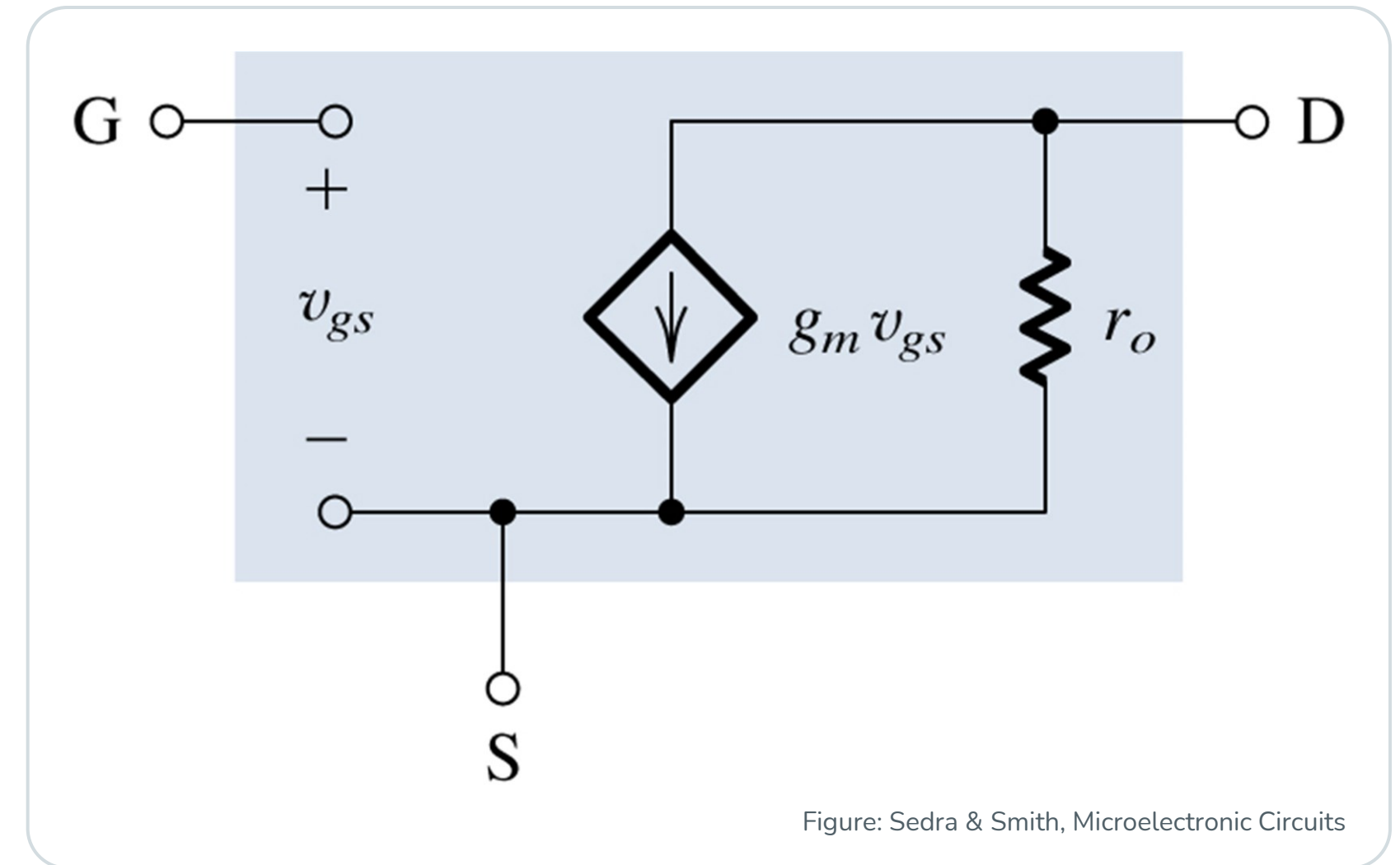
- 4 With a drain resistor R_D , the voltage gain is

$$A_v = \frac{v_{ds}}{v_{gs}} = -g_m R_D$$

Small-signal equivalent circuits



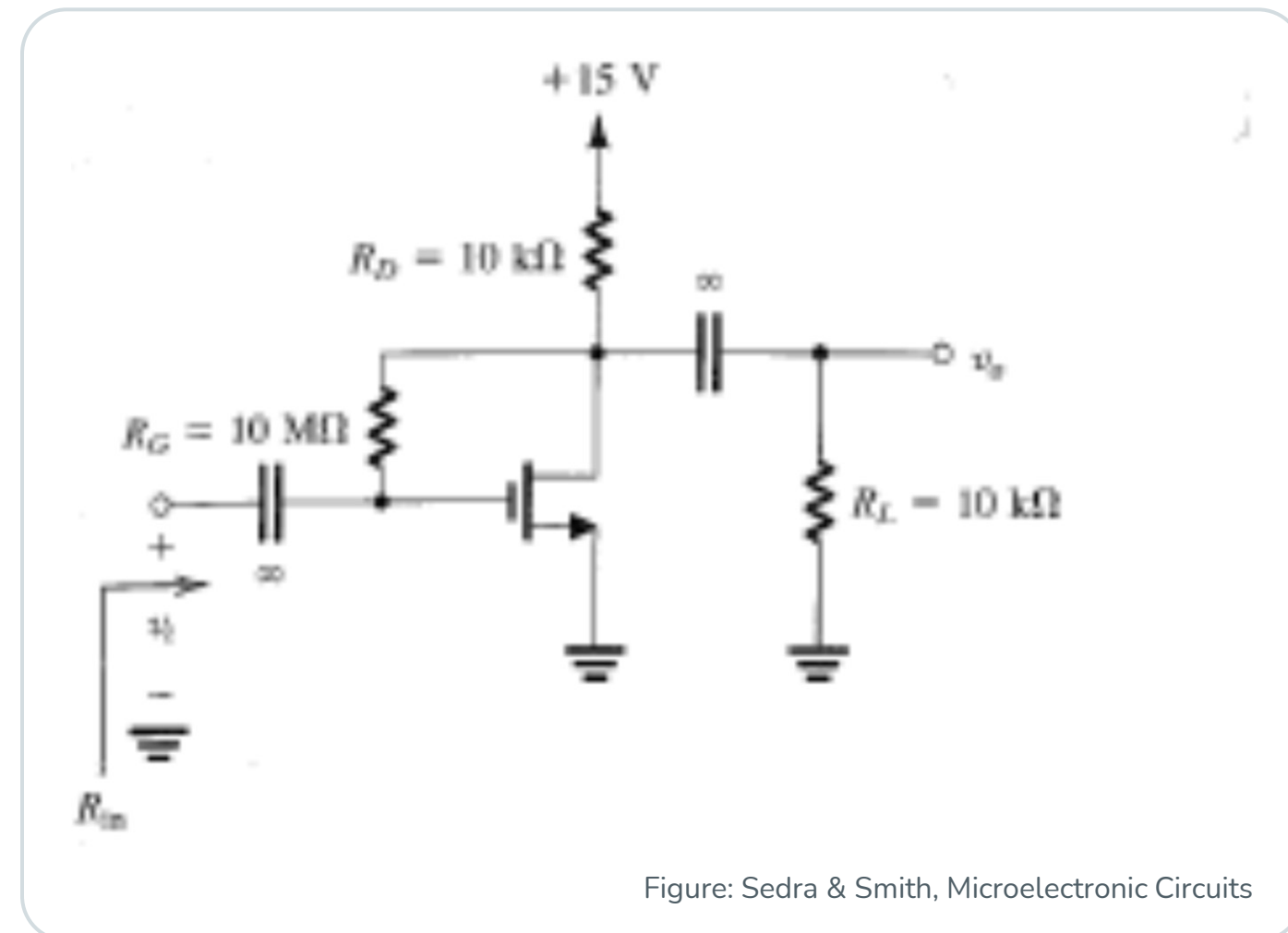
- The MOSFET as a voltage-controlled current source $g_m v_{gs}$; no current into the gate.



- Adding $r_o = V_A/I_D$ models the slight slope of the saturation curves (V_A : the Early voltage).

Example: gain and input resistance

$V_t = 1.5 \text{ V}$, $k'_n(W/L) = 0.25 \text{ mA/V}^2$, $V_A = 50 \text{ V}$. Coupling capacitors are short circuits for the signal. Find $A_v = v_o/v_i$ and R_{in} .



- 1 DC: no gate current, so $V_{GS} = V_D$ (I in mA)

$$V_D = 15 - 10I_D, \quad I_D = \frac{1}{2}(0.25)(V_D - 1.5)^2$$

$$I_D = 1.06 \text{ mA}, \quad V_{GS} = V_D = 4.4 \text{ V}$$
- 2 Small-signal parameters

$$g_m = 0.25(4.4 - 1.5) = 0.728 \text{ mA/V}, \quad r_o = \frac{50}{1.06} = 47 \text{ k}\Omega$$
- 3 Gain (R_G carries negligible signal current)

$$A_v \approx -g_m(R_D \parallel R_L \parallel r_o) = -0.728(4.52) = -3.29 \text{ V/V}$$
- 4 Input resistance: R_G bridges input and output (Miller effect)

$$R_{in} = \frac{R_G}{1 - A_v} = \frac{10}{4.29} = 2.33 \text{ M}\Omega$$

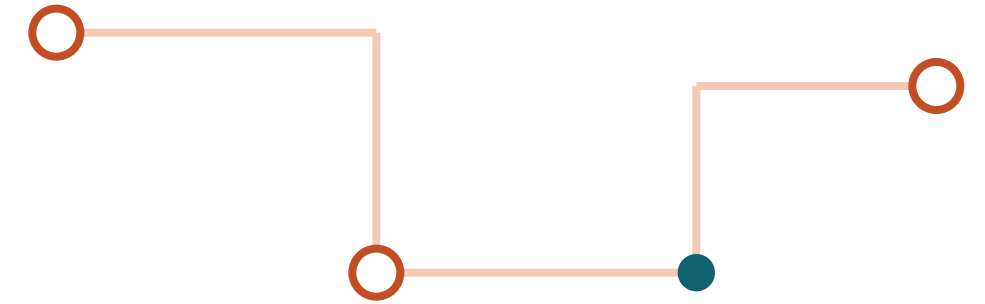
$$A_v \approx -3.3 \text{ V/V}$$

$$R_{in} = 2.33 \text{ M}\Omega$$

5

The MOSFET as a switch

- Cutoff: an open switch
- Deep triode: a closed switch



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The MOSFET as a switch

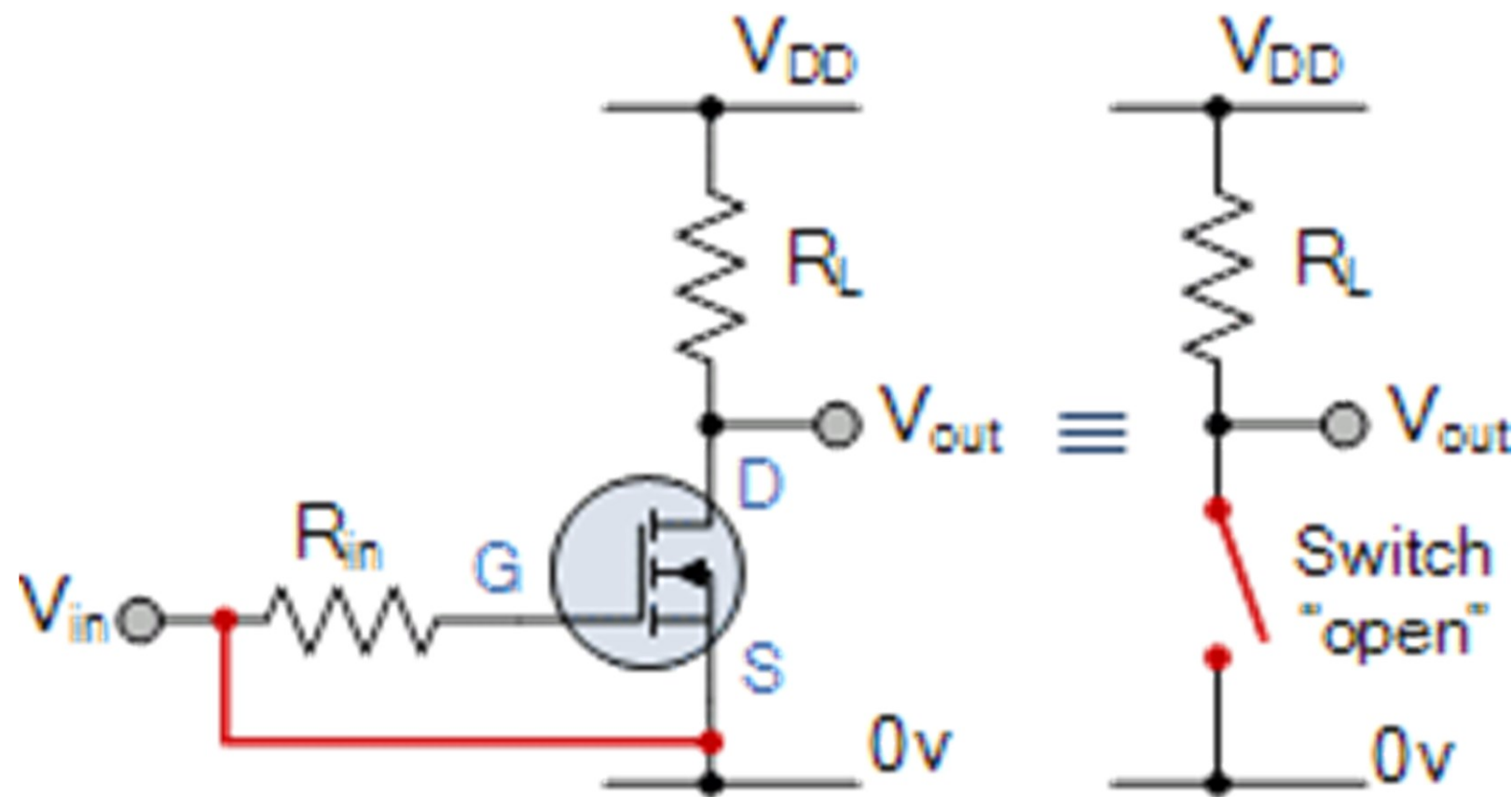


Figure: web source, via the original lecture slides

- $v_{GS} < V_t$: **cutoff**. The switch is **open** and $i_D = 0$.
- v_{GS} high and v_{DS} small: deep **triode**. The switch is **closed**, with a small resistance r_{DS} .
- Driving the gate between these two states is the basis of digital logic.

Summary: three regions of operation

Region	Condition	Drain current
Cutoff	$v_{GS} \leq V_t$	$i_D = 0$
Triode	$v_{GS} > V_t, v_{DS} < v_{GS} - V_t$	$i_D = k'_n \frac{W}{L} \left[(v_{GS} - V_t)v_{DS} - \frac{1}{2}v_{DS}^2 \right]$
Saturation	$v_{GS} > V_t, v_{DS} \geq v_{GS} - V_t$	$i_D = \frac{1}{2}k'_n \frac{W}{L} (v_{GS} - V_t)^2$

- DC problems: **assume** a region, **solve**, then **check** the assumption.
- Small signals: $g_m = k'_n (W/L)(V_{GS} - V_t)$; a common-source stage has $A_v = -g_m R_D$.

Thank you

Next lecture

Lecture 4: Bipolar junction transistors (BJTs)

With thanks to **Dr. Sameh Osama Ezzat**, who prepared the original material for this lecture.

Figures: Sedra & Smith, Microelectronic Circuits (Oxford University Press); other diagrams from web sources via the original slides.

